ISM · ASSIGNMENT - I

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[redacted]

SECTION - 7

1) Number of defective items produced per month : 12, 15, 14, 13, 16, 18, 14, 15, 13, 14, 17, 14, 36

a) Mean, Mode and median

$n = 13$

$\Sigma x_i = 12 + 15 + 14 + 13 + 16 + 18 + 14 + 15 + 13 + 14 + 17 + 14 + 36$

$= 211$

$\bar{x} = \dfrac{\Sigma x_i}{n} = 211/13 = 16.23$

mean = 16.23

median:

Sorted data:

12, 13, 13, 14, 14, 14, 14, 15, 15, 16, 17, 18, 36

Position = $\dfrac{n+1}{2}$ = 7^{th} position

median = 14

mode

	Frequency
12	1
13	2
14	4
15	2
16	1
17	1
18	1
36	1

mode = 14

mean > median > mode

So, the distribution is positively skewed (right)

b) $X_{min} = 12$

$Q_1 : (n+1)/4 = 3.5$

$Q_1 = \frac{X_3 + X_4}{2} = 27/2 = 13.5$

$Q_2 = \text{median} = 14$

$Q_3 : \frac{3(n+1)}{4} = 10.5$

$Q_3 = \frac{X_{10} + X_{11}}{2} = 31/2 = 15.5$

$X_{max} = 36$

5 point summary: (12, 13.5, 14, 15.5, 36)

$IQR = Q_3 - Q_1 = 2$

Mild outlier boundary: $Q_1 - 1.5\,IQR$ to $Q_3 + 1.5\,IQR$

$= 10.5$ to 18.5

Mild outlier : 36

Extreme outlier boundary:

$Q_1 - 3\,IQR$ to $Q_3 + 3\,IQR$

$= 7.5$ to 21.5

Extreme outlier : 36

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c) Mean without outlier

$= 175/12 = 14.58$

$\bar{X}_{13} - \bar{X}_{12} = 16.23 - 14.58 = 1.65$

percentage increase $= \frac{1.65}{14.58} \times 100 = 11.3\%$

Median without outlier $= \frac{X_6 + X_7}{2} = 14$

Median remains the same.

S.D

$$\sigma_{13} = \sqrt{\frac{\Sigma(x-\bar{x}_{13})^2}{n}} = \sqrt{\frac{\Sigma(x-\bar{x}_{13})^2}{13}} = 6.47$$

$CV_{13} = \sigma/\bar{x} \times 100 = 39.9\%$

SD without outlier

$$\sigma_{12} = \sqrt{\frac{\Sigma(x-\bar{x}_{12})^2}{12}} = 1.73$$

$CV_{12} = \frac{1.73}{14.58} \times 100 = 11.9\%$

Since, Coefficient of variation and mean is highly sensitive to outlier, Median is more justified to be reported

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a)

Visitor	Device	Subscribe
1	Mobile	No
2	Mobile	Yes
3	Desktop	Yes
4	Desktop	No
5	Tablet	Yes

a) $n(\text{Yes}) = 3$

$n(\text{No}) = 2$

$$P(\text{sub} = \text{yes}) = \frac{n(\text{Yes})}{n(\text{Tot})} = 3/5 = 0.6$$

$$P(\text{sub} = \text{No}) = \frac{n(\text{no})}{n(\text{Tot})} = 2/5 = 0.4$$

b) $P(\text{Yes} \mid \text{Mobile}) = P(\text{yes}) \times P(\text{Mobile} \mid \text{yes})$

$= 3/5 \times 1/3 = 0.2$

$P(\text{No} \mid \text{Mobile}) = P(\text{No}) \times P(\text{mobile} \mid \text{No})$

$= 2/5 \times 1/2 = 0.2$

so customer is equally likely to subscribe

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3. $n(Total) = 500$

$n(D) = 230 \quad n(C) = 180 \quad n(B) = 150$

$n(D \cap C) = 90 \quad n(C \cap B) = 60 \quad n(D \cap B) = 70$

$n(D \cap C \cap B) = 35$

1) $n(D \cup C \cup B) = n(D) + n(C) + n(B) - (n(D \cap C) + n(D \cap B) + n(C \cap B)) + n(D \cap C \cap B)$

$= 230 + 180 + 150 - 90 - 70 - 60 + 35$

$= 375$

$P(\text{atleast one course}) = \dfrac{n(D \cup C \cup B)}{n(Total)}$

$= 375/500 = 0.75$

2) $P(\text{~~atle~~ exactly 2 courses}) = [n(D \cap C) + n(D \cap B) + n(C \cap B) - 3n(D \cap B \cap C)]/n(Total)$

$= (90 + 70 + 60 - 105)/500$

$= 115/500 = 0.23$

3) $P(C|D) = \dfrac{n(C \cap D)}{n(D)} = 90/230 = 0.39$

4) $P(D) \times P(C) = 0.46 \times 0.36 = 0.17$

$P(D \cap C) = 90/500 = 0.18$

$P(D) \times P(C) \neq P(D \cap C) \quad \therefore$ Independent

5) From 3,

39% of students enrolled in Data science prefer CyberSecurity